FURTHER MECHANICS / MISCELLANEOUS MECHANICS

# Dimensional Analysis

Worked solutions 10 questions · 48 marks

**Instructions**

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of $g$ is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

## Questions


| | | |
|---|---|---|
| Question 1 | *5 marks* | 2 |
| Question 2 | *5 marks* | 4 |
| Question 3 | *6 marks* | 6 |
| Question 4 | *4 marks* | 8 |
| Question 5 | *6 marks* | 10 |
| Question 6 | *6 marks* | 12 |
| Question 7 | *2 marks* | 14 |
| Question 8 | *4 marks* | 15 |
| Question 9 | *6 marks* | 16 |
| Question 10 | *4 marks* | 18 |

**1.** **(a)** State the dimensions of force. **[1]**

**(b)** The braking distance of a vehicle can be modelled by the formula

$$d = m^a u^b R^c$$

where,

$m$ is the mass of the vehicle in kg
$u$ is the initial speed in $\mathrm{m\,s^{-1}}$
$R$ is the constant braking force in N

Use dimensional analysis to find the values of $a$, $b$ and $c$. **[4]**

**Solution**

**(a)** Force is mass $\times$ acceleration, so

$$[F] = M \times LT^{-2} = MLT^{-2}$$

**(b)** Write the dimensions of each quantity:

$$[d] = L, \qquad [m] = M, \qquad [u] = LT^{-1}, \qquad [R] = MLT^{-2}$$

Given

$$d = m^a u^b R^c$$

the dimensions on the right-hand side are

$$\begin{aligned}[m^a u^b R^c] &= M^a (LT^{-1})^b (MLT^{-2})^c \\ &= M^a L^b T^{-b} \times M^c L^c T^{-2c} \\ &= M^{a+c} L^{b+c} T^{-b-2c}\end{aligned}$$

Since this must equal the dimensions of $d$,

$$M^{a+c} L^{b+c} T^{-b-2c} = L = M^0 L^1 T^0$$

Equating powers of $M$, $L$ and $T$ gives

$$a + c = 0, \qquad b + c = 1, \qquad -b - 2c = 0$$

From $-b - 2c = 0$,

$$b = -2c$$

Substitute into $b + c = 1$:

$$-2c + c = 1$$

$$-c = 1$$

$$c = -1$$

Then

$$b = -2(-1) = 2$$

and from $a + c = 0$,

$$a + (-1) = 0$$

so

$$a = 1$$

Hence

$$a = 1, \qquad b = 2, \qquad c = -1$$

**2.** A small satellite moves in a circular orbit of radius $r$ about the centre of a much more massive planet of mass $M$. It is suggested that the period $T$ of the orbit depends only on $G$, $M$ and $r$, and that

$$T = kG^{\alpha}M^{\beta}r^{\gamma}$$

where $k$ is a dimensionless constant.

It is given that

$$G = 6.67 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$$

**(a)** Write down the dimensions of $G$. **[1]**

**(b)** By using dimensional analysis, determine the values of $\alpha$, $\beta$ and $\gamma$. **[4]**

**Solution**

**(a)** From

$$G = 6.67 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$$

the dimensions of $G$ are

$$[G] = M^{-1}L^3T^{-2}$$

**(b)** Since $k$ is dimensionless,

$$[T] = [G]^{\alpha}[M]^{\beta}[r]^{\gamma}$$

Now

$$[T] = T, \qquad [G] = M^{-1}L^3T^{-2}, \qquad [r] = L$$

so

$$\begin{aligned} T &= (M^{-1}L^3T^{-2})^{\alpha}M^{\beta}L^{\gamma} \\ &= M^{-\alpha+\beta}L^{3\alpha+\gamma}T^{-2\alpha} \end{aligned}$$

Equating powers of $M$, $L$ and $T$ with $M^0L^0T^1$:

$$\begin{aligned} -\alpha + \beta &= 0 \\ 3\alpha + \gamma &= 0 \\ -2\alpha &= 1 \end{aligned}$$

From $-2\alpha = 1$,

$$\alpha = -\frac{1}{2}$$

Then

$$\beta = \alpha = -\frac{1}{2}$$

and

$$\gamma = -3\alpha = \frac{3}{2}$$

Hence

$$\alpha = -\frac{1}{2}, \qquad \beta = -\frac{1}{2}, \qquad \gamma = \frac{3}{2}$$

**3.** A small probe is launched vertically upwards from the surface of a spherical moon. The radius of the moon is $R$ metres, and the probe is at a distance $r$ metres from the centre of the moon. The gravitational field strength $g$ at distance $r$ is modelled by

$$g = \frac{\Gamma}{r^2}$$

where $\Gamma$ is a constant associated with the moon.

The probe has initial speed $u$ at the surface and speed $v$ when it reaches distance $r$. These speeds are related by

$$v^2 = u^2 - 2\Gamma\left(\frac{1}{R} - \frac{1}{r}\right)$$

The work done against gravity in moving a probe of mass $m$ from the surface to distance $r$ is

$$W = m\Gamma\left(\frac{1}{R} - \frac{1}{r}\right)$$

**(a)** Find the dimensions of $\Gamma$. **[2]**

**(b)** Use dimensional analysis to show that the equation for $v^2$ is dimensionally consistent. **[2]**

**(c)** Find the dimensions and SI units of $W$. **[2]**

**Solution**

**(a)** From

$$g = \frac{\Gamma}{r^2}$$

and since

$$[g] = LT^{-2}, \qquad [r] = L$$

we have

$$[\Gamma] = [g][r^2] = LT^{-2} \times L^2 = L^3T^{-2}$$

**(b)** In

$$v^2 = u^2 - 2\Gamma\left(\frac{1}{R} - \frac{1}{r}\right)$$

the dimensions of $v^2$ and $u^2$ are

$$[v^2] = [u^2] = (LT^{-1})^2 = L^2T^{-2}$$

Also,

$$\left[\frac{1}{R}\right] = \left[\frac{1}{r}\right] = L^{-1}$$

so

$$\left[\left(\frac{1}{R} - \frac{1}{r}\right)\right] = L^{-1}$$

Since 2 is dimensionless,

$$\left[2\Gamma\left(\frac{1}{R}-\frac{1}{r}\right)\right] = [\Gamma]L^{-1} = L^3T^{-2}\times L^{-1} = L^2T^{-2}$$

Therefore every term in the equation has dimensions $L^2T^{-2}$, so the equation is dimensionally consistent.

**(c)** Using

$$W = m\Gamma\left(\frac{1}{R}-\frac{1}{r}\right)$$

with

$$[m] = M, \qquad [\Gamma] = L^3T^{-2}, \qquad \left[\left(\frac{1}{R}-\frac{1}{r}\right)\right] = L^{-1}$$

gives

$$[W] = M\times L^3T^{-2}\times L^{-1} = ML^2T^{-2}$$

Its SI units are

$$\mathrm{kg\,m^2\,s^{-2}}$$

which is the joule, J.

**4.** The increase in pressure $p$ pascals between the surface and the bottom of a liquid of depth $h$ metres and density $\rho\,\mathrm{kg\,m^{-3}}$ is modelled by

$$p = k\rho^{\alpha}h^{\beta}g$$

where $g\,\mathrm{m\,s^{-2}}$ is the acceleration due to gravity and $k$ is a dimensionless constant.

**(a)** State the dimensions of $p$. **[1]**

**(b)** Use dimensional analysis to find the value of $\alpha$ and the value of $\beta$. **[3]**

**Solution**

**(a)** Pressure is force per unit area.

Since [force] $= MLT^{-2}$ and [area] $= L^2$,

$$[p] = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$$

**(b)** Using

$$p = k\rho^{\alpha}h^{\beta}g$$

and noting that $k$ is dimensionless,

$$[p] = [\rho]^{\alpha}[h]^{\beta}[g]$$

Now

$$[\rho] = ML^{-3}, \qquad [h] = L, \qquad [g] = LT^{-2}$$

So the dimensions of the right-hand side are

$$\begin{aligned}(ML^{-3})^{\alpha}(L)^{\beta}(LT^{-2}) &= M^{\alpha}L^{-3\alpha}L^{\beta}LT^{-2}\\ &= M^{\alpha}L^{-3\alpha+\beta+1}T^{-2}\end{aligned}$$

But from part $(a)$,

$$[p] = ML^{-1}T^{-2}$$

Equating powers of $M$, $L$ and $T$:

$$\alpha = 1$$

and

$$-3\alpha + \beta + 1 = -1$$

Substituting $\alpha = 1$,

$$\begin{aligned}-3(1) + \beta + 1 &= -1\\ \beta - 2 &= -1\\ \beta &= 1\end{aligned}$$

Hence

$$\alpha = 1, \qquad \beta = 1$$

**5.** The force, $F$, exerted on a flat plate by a water jet is thought to depend on the density of the water, $\rho$, the cross-sectional area of the jet, $A$, and the speed of the water, $v$. You are given that $\rho$ is measured in $\text{kg m}^{-3}$.

**(a)** Find the dimensions of density $\rho$. **[1]**

A model of the force suggests the relationship

$$F = \frac{1}{2}\rho^{\alpha} A^{\beta} v^{\gamma}$$

**(b)** Explain whether or not it is possible to use dimensional analysis to verify that the constant $\frac{1}{2}$ is correct. **[1]**

**(c)** Use dimensional analysis to find the values of $\alpha$, $\beta$ and $\gamma$. **[4]**

**Solution**

**(a)** Density is mass per unit volume.

$$[\rho] = \frac{M}{L^3} = ML^{-3}$$

**(b)** No, it is not possible.

Dimensional analysis only checks that both sides of an equation have the same dimensions. The number $\frac{1}{2}$ is a dimensionless constant, so dimensional analysis cannot show whether $\frac{1}{2}$ is the correct numerical value.

**(c)** We use

$$F = \frac{1}{2}\rho^{\alpha} A^{\beta} v^{\gamma}$$

The factor $\frac{1}{2}$ is dimensionless, so it does not affect the dimensions.

Now write the dimensions of each quantity:

$$[F] = MLT^{-2}, \qquad [\rho] = ML^{-3}, \qquad [A] = L^2, \qquad [v] = LT^{-1}$$

So

$$MLT^{-2} = (ML^{-3})^{\alpha}(L^2)^{\beta}(LT^{-1})^{\gamma}$$

Combining powers gives

$$MLT^{-2} = M^{\alpha} L^{-3\alpha+2\beta+\gamma} T^{-\gamma}$$

Equating powers of $M$, $L$ and $T$:

$$\text{For } M: \quad \alpha = 1$$

$$\text{For } T: \quad -\gamma = -2 \Rightarrow \gamma = 2$$

$$\text{For } L: \quad -3\alpha + 2\beta + \gamma = 1$$

Substitute $\alpha = 1$ and $\gamma = 2$ into the $L$-equation:

$$\begin{aligned} -3(1) + 2\beta + 2 &= 1 \\ -1 + 2\beta &= 1 \\ 2\beta &= 2 \\ \beta &= 1 \end{aligned}$$

Hence

$$\alpha = 1, \qquad \beta = 1, \qquad \gamma = 2$$

So the model is

$$F = \frac{1}{2}\rho A v^2$$

**6.** A motor pulls a crate of mass $m$ along a rough horizontal floor. The equation below is suggested to model the pulling force exerted by the motor, $F$, while the crate moves in a straight line.

$$F = k_1 mv\frac{\mathrm{d}v}{\mathrm{d}x} + k_2\mu mg + k_3\frac{Q}{x}$$

in which,

- $v$ is the speed of the crate when it has travelled distance $x$
- $\mu$ is the coefficient of friction between the crate and the floor
- $g$ is the acceleration due to gravity
- $Q$ is the total thermal energy transferred to the surroundings after the crate has moved distance $x$
- $k_1$, $k_2$ and $k_3$ are dimensionless constants

Determine whether the equation is dimensionally consistent. **[6]**

**Solution**

For the equation to be dimensionally consistent, every term must have the dimensions of force.

$$[F] = MLT^{-2}$$

Also, $k_1$, $k_2$ and $k_3$ are dimensionless, and $\mu$ is dimensionless.

Now consider each term on the right-hand side.

$$[v] = LT^{-1}, \qquad [x] = L$$

so

$$\left[\frac{\mathrm{d}v}{\mathrm{d}x}\right] = \frac{[v]}{[x]} = \frac{LT^{-1}}{L} = T^{-1}$$

Hence for the first term,

$$\begin{aligned}\left[k_1 mv\frac{\mathrm{d}v}{\mathrm{d}x}\right] &= [m][v]\left[\frac{\mathrm{d}v}{\mathrm{d}x}\right] \\ &= M \cdot LT^{-1} \cdot T^{-1} \\ &= MLT^{-2}\end{aligned}$$

For the second term,

$$\begin{aligned}[k_2\mu mg] &= [\mu][m][g] \\ &= 1 \cdot M \cdot LT^{-2} \\ &= MLT^{-2}\end{aligned}$$

For the third term, $Q$ is thermal energy transferred, so it has the dimensions of energy:

$$[Q] = ML^2T^{-2}$$

Therefore

$$\begin{aligned}\left[k_3\frac{Q}{x}\right] &= \frac{[Q]}{[x]} \\ &= \frac{ML^2T^{-2}}{L} \\ &= MLT^{-2}\end{aligned}$$

Each term on the right-hand side has dimensions $MLT^{-2}$, the same as force.

Therefore the equation is dimensionally consistent.

**7.** The speed of a particle falling vertically through a resistive medium is modelled by the formula

$$v = \frac{mg}{k}\left(1 - e^{-kt/m}\right)$$

where,

$v$ is the speed of the particle
$m$ is the mass of the particle
$g$ is the acceleration due to gravity
$t$ is the time since the particle was released
$k$ is a constant

Use dimensional analysis to determine the dimensions of $k$. **[2]**

**Solution**

For

$$v = \frac{mg}{k}\left(1 - e^{-kt/m}\right)$$

the factor $1 - e^{-kt/m}$ must be dimensionless.

In particular, the exponent of an exponential must have no dimensions, so

$$\left[\frac{kt}{m}\right] = 1$$

Using

$$[t] = T, \qquad [m] = M$$

we get

$$\frac{[k][t]}{[m]} = 1$$

so

$$\frac{[k]T}{M} = 1$$

Hence

$$[k] = MT^{-1}$$

As a check,

$$\left[\frac{mg}{k}\right] = \frac{M \cdot LT^{-2}}{MT^{-1}} = LT^{-1}$$

which is the dimension of speed, as required.

Therefore $k$ has dimensions $MT^{-1}$, with SI unit $\mathrm{kg\,s^{-1}}$.

**8.** A comet of mass $m$ measured in kg follows an elliptical orbit around a planet. The semi-major axis of the orbit is $a$ measured in metres and the orbital period is $T$ measured in seconds. The eccentricity $e$ of the orbit is dimensionless.

A constant $\mu$ associated with the planet is defined by

$$T^2 = \frac{4\pi^2 a^3}{\mu}$$

and the angular momentum of the comet is given by

$$H = m\sqrt{\mu a(1-e^2)}$$

By using dimensional analysis, find the dimensions of:

**(a)** $\mu$ **[2]**

**(b)** $H$ **[2]**

**Solution**

**(a)** From

$$T^2 = \frac{4\pi^2 a^3}{\mu}$$

and since $4\pi^2$ is dimensionless,

$$[T]^2 = \frac{[a]^3}{[\mu]}$$

Using $[T] = T$ and $[a] = L$,

$$T^2 = \frac{L^3}{[\mu]}$$

so

$$[\mu] = \frac{L^3}{T^2} = L^3T^{-2}$$

**(b)** The angular momentum is

$$H = m\sqrt{\mu a(1-e^2)}$$

Here $[m] = M$, and since $e$ is dimensionless, $(1-e^2)$ is also dimensionless.

Therefore

$$[H] = [m]\sqrt{[\mu][a]}$$

Using $[\mu] = L^3T^{-2}$ from part (a) and $[a] = L$,

$$\begin{aligned}[H] &= M\sqrt{(L^3T^{-2})(L)} \\ &= M\sqrt{L^4T^{-2}} \\ &= ML^2T^{-1}\end{aligned}$$

**9.** A small ball of mass $m$ kg is projected vertically upwards with speed $u\,\mathrm{m\,s^{-1}}$. During the motion, when its speed is $v\,\mathrm{m\,s^{-1}}$, it experiences an air resistance of magnitude $kv^2$ N, where $k$ is a constant.

**(a)** Determine the dimensions of $k$. **[2]**

The greatest height reached by the ball is $H$ m. A formula approximating $H$ is

$$H = \frac{u^2}{2g} - \frac{ku^4}{4mg^2} + \frac{1}{6}k^2m^\alpha u^\beta g^\gamma$$

where $g\,\mathrm{m\,s^{-2}}$ is the acceleration due to gravity.

**(b)** Use dimensional analysis to find $\alpha$, $\beta$ and $\gamma$. **[4]**

**Solution**

**(a)** The air resistance has magnitude $kv^2$, and this is a force.

$$[k][v]^2 = [F]$$

Now

$$[F] = MLT^{-2}, \qquad [v] = LT^{-1}$$

So

$$\begin{aligned}[k] &= \frac{[F]}{[v]^2} \\ &= \frac{MLT^{-2}}{(LT^{-1})^2} \\ &= \frac{MLT^{-2}}{L^2T^{-2}} \\ &= ML^{-1}\end{aligned}$$

Hence the SI unit of $k$ is $\mathrm{kg\,m^{-1}}$.

**(b)** Since $H$ is a height,

$$[H] = L$$

Therefore each term in

$$H = \frac{u^2}{2g} - \frac{ku^4}{4mg^2} + \frac{1}{6}k^2m^\alpha u^\beta g^\gamma$$

must have dimensions of length.

So the third term must satisfy

$$[k^2m^\alpha u^\beta g^\gamma] = L$$

Using

$$[k] = ML^{-1}, \qquad [m] = M, \qquad [u] = LT^{-1}, \qquad [g] = LT^{-2}$$

we get

$$\begin{aligned}[k^2 m^\alpha u^\beta g^\gamma] &= (ML^{-1})^2 (M)^\alpha (LT^{-1})^\beta (LT^{-2})^\gamma \\ &= M^{2+\alpha} L^{-2+\beta+\gamma} T^{-\beta-2\gamma}\end{aligned}$$

This must equal $M^0 L^1 T^0$, so equating powers gives

$$\begin{aligned}2 + \alpha &= 0 \\ -2 + \beta + \gamma &= 1 \\ -\beta - 2\gamma &= 0\end{aligned}$$

From the first equation,

$$\alpha = -2$$

From the third equation,

$$\beta = -2\gamma$$

Substitute into the second equation:

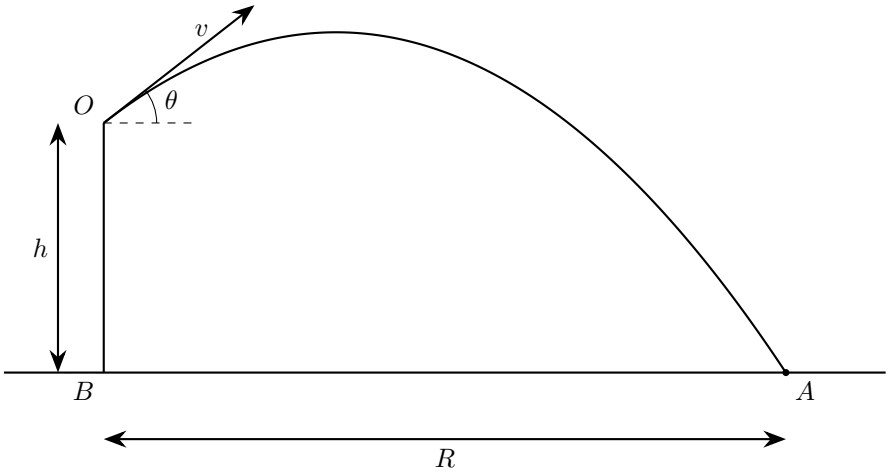
$$\begin{aligned}-2 + (-2\gamma) + \gamma &= 1 \\ -2 - \gamma &= 1 \\ -\gamma &= 3 \\ \gamma &= -3\end{aligned}$$

Then

$$\beta = -2(-3) = 6$$

Hence

$$\alpha = -2, \qquad \beta = 6, \qquad \gamma = -3$$




**10.** A particle is projected with speed $v$ from a point $O$ at the top of a cliff of height $h$ above horizontal ground. The angle of projection is $\theta$ above the horizontal. The particle lands at the point $A$ on the ground, and $B$ is the foot of the cliff, so that $BA = R$, as shown in the diagram.

You are given that

$$R = \frac{v^\alpha \cos\theta}{g^\beta}\left(v^\gamma \sin\theta + \sqrt{v^\delta \sin^2\theta + 2gh}\right)$$

Use dimensional analysis to find $\alpha$, $\beta$, $\gamma$ and $\delta$. **[4]**

**Solution**

Using dimensions,

$$[R] = L, \qquad [v] = LT^{-1}, \qquad [g] = LT^{-2}, \qquad [h] = L$$

and $\sin\theta$ and $\cos\theta$ are dimensionless.

First look at the expression inside the square root. Since terms being added must have the same dimensions,

$$[v^\delta \sin^2\theta] = [2gh]$$

Now

$$[2gh] = [g][h] = (LT^{-2})(L) = L^2T^{-2}$$

So

$$(LT^{-1})^\delta = L^2T^{-2}$$

Comparing powers of $L$ and $T$,

$$\delta = 2$$

Hence the whole square-root term has dimensions

$$\sqrt{L^2T^{-2}} = LT^{-1}$$

Therefore $v^\gamma \sin\theta$ must also have dimensions $LT^{-1}$, so

$$(LT^{-1})^\gamma = LT^{-1}$$

which gives

$$\gamma = 1$$

Now the bracket

$$v^\gamma \sin\theta + \sqrt{v^\delta \sin^2\theta + 2gh}$$

has dimensions $LT^{-1}$. So

$$[R] = \left[\frac{v^\alpha \cos\theta}{g^\beta}\right]\left[v^\gamma \sin\theta + \sqrt{v^\delta \sin^2\theta + 2gh}\right]$$

$$L = \frac{(LT^{-1})^\alpha}{(LT^{-2})^\beta}(LT^{-1})$$

$$= L^{\alpha-\beta+1}T^{-\alpha+2\beta-1}$$

Comparing powers with $L^1T^0$,

$$\alpha - \beta + 1 = 1, \qquad -\alpha + 2\beta - 1 = 0$$

From the first equation,

$$\alpha = \beta$$

Substitute into the second:

$$-\beta + 2\beta - 1 = 0$$

so

$$\beta = 1$$

and therefore

$$\alpha = 1$$

So the values are

$$\alpha = 1, \qquad \beta = 1, \qquad \gamma = 1, \qquad \delta = 2$$